



King Abdulaziz University

Faculty of Computing & Information Technology Computer Science Department

Exam 1 (June 27th, 2019)
Course: CPCS-223 (Analysis and Design of Algorithms)
Term: Summer 2019
Points: 60 Time: 80 minutes

SOLUTION

Student ID: _____

Student Name: _____

*****Section:** _____ **AA**

Q	SO	Question	Max. Marks	Obt. Marks
01	A	1	10	
02	A	2	10	
03	A	3	10	
04	B	4	10	
05	B	5	10	
06	A	6	10	

Instructor:

☐ Dr Muhammad Umair

CLO – 1
Question No. 1

SO – A
(05+ 05) points

PART – A

Write down at least 05 characteristics of an “Algorithm”.

Solution:

S. No.	Characteristics
1	It accepts one or more inputs
2	It returns at least one output
3	It terminates after finite steps
4	It is independent of a specific programming language, machine or compiler
5	for every input it halts with correct output

PART – B

What is “Problem Statement”. Discuss its main ingredients briefly. Also mention the role of an “Algorithm” in problem solving.

Solution:

Problem Statement:

(01) pt

The statement of a given problem which defines the relationship between the input and output

Ingredients of Problem Statement:

(03) pts

Input

The values received from user

It is optional to mention in problem statement

Process

Making calculations or manipulating the input

It is optional to mention in problem statement

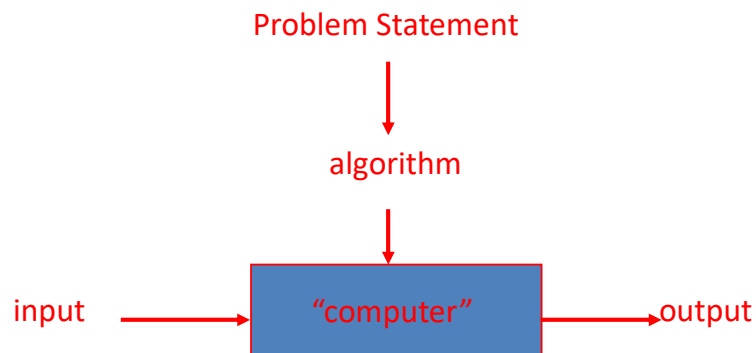
Output

It is the outcome of the program to the user

It must be mentioned in problem statement

Role of Algorithm:

(01) pt



Question No. 2**10 points**

On which parameters the efficiency of an algorithm depends? Describe the main characteristics of “Time Efficiency” for an algorithm.

Solution:Efficiency Parameters:*(02) pts***1) Speed;**

to determine the performance of the algorithm in terms of *time taken* (Time Efficiency)

2) Memory;

to determine the performance of the algorithm in terms of storage requirements (Space Efficiency)

Time Efficiency:*(08) pts*

S. No.	Characteristics
1	Determines how fast an algorithm solves a given problem.
2	It is the measure of time taken by the algorithm to produce the output with a given input.
3	Assuming that an algorithm takes $T(n)$ time to process n data items. The function $T(n)$ is referred to as running time of the algorithm.
4	It is also known as time complexity.
	The time complexity can depend on more than one parameter. For example, the running time of a graph algorithm can depend on both the number of vertices and edges in the graph.

CLO – 3
Question No. 3

SO – A
(03+ 07) points

PART – A

For the given recurrence; find out the upper bound (O, or o) with which we may approximate the running time.

$$f(n) = 3T(\lfloor n/4 \rfloor) + cn^2$$

Solution:

$$\begin{aligned} T(n) &\leq 3T(\lfloor n/4 \rfloor) + cn^2 \\ &\leq 3d \lfloor n/4 \rfloor^2 + cn^2 \\ &\leq 3d(n/4)^2 + cn^2 \\ &= \frac{3}{16} dn^2 + cn^2 \\ &\leq dn^2, \quad d \geq (16/13)c. \end{aligned}$$

PART – B

Solve the given recurrence using “Substitution Method”

$$T(n) = 2T(\lfloor n/2 \rfloor + 17) + n$$

Solution:

Guess the Solution:

(01) pt

Prove by Mathematical Induction:

(01 + 04 + 01) pts

Assumption:

Prove for $T(n)$

Boundary Conditions:

(01) pt

CLO – 4

SO – B

Question No. 4

10 points

Find the running time of the following algorithm using the RAM model and approximate the running time with Θ – notation.

```
ALGORITHM Question04 ( $A[0..n-1], l, u$ )  
  for  $j \leftarrow 0$  to  $u-l$  do  $D[j] \leftarrow 0$   
  for  $i \leftarrow 0$  to  $n-1$  do  $D[A[i]-l] \leftarrow D[A[i]-l] + 1$   
  for  $j \leftarrow 1$  to  $u-l$  do  $D[j] \leftarrow D[j-1] + D[j]$   
  for  $i \leftarrow n-1$  downto  $0$  do  
     $j \leftarrow A[i]-l$   
     $S[D[j]-1] \leftarrow A[i]$   
     $D[j] \leftarrow D[j]-1$   
  return  $S$ 
```

Solution:

(08 pts)

Statement #	Cost	Times(# of Iterations)
<u>1</u>		
<u>2</u>		
<u>3</u>		
<u>4</u>		
<u>5</u>		
<u>6</u>		
<u>7</u>		
<u>8</u>		

$$\mathbf{T}(\mathbf{n}) =$$

(02 pts)

CLO – 5**Question No. 5**

For the given data set, how the arrays A, count and S will be updated during execution of the given algorithm of “Question05”? You are required to reflect all changes to these arrays for each iteration.

SO – B**10 points****ALGORITHM** *Question05* ($A[0..n-1]$)

```

for  $i \leftarrow 0$  to  $n-1$  do  $Count[i] \leftarrow 0$ 
for  $i \leftarrow 0$  to  $n-2$  do
    for  $j \leftarrow i+1$  to  $n-1$  do
        if  $A[i] < A[j]$ 
             $Count[j] \leftarrow Count[j] + 1$ 
        else  $Count[i] \leftarrow Count[i] + 1$ 
for  $i \leftarrow 0$  to  $n-1$  do  $S[Count[i]] \leftarrow A[i]$ 
return  $S$ 

```

Solution:**A**

0	1	2	3	4	5	6	7	8	9
49	68	05	61	35	57	6	44	99	91

Count

Iteration	0	1	2	3	4	5	6	7	8	9
1	4	1	0	1	0	1	0	0	1	1
2	4	7	0	1	0	1	0	0	2	2
3	4	7	0	2	1	2	1	1	3	3
4	4	7	0	6	1	2	1	1	4	4
5	4	7	0	6	2	3	1	2	5	5
6	4	7	0	6	2	5	1	2	6	6
7	4	7	0	6	2	5	1	3	7	7
8	4	7	0	6	2	5	1	3	8	8
9	4	7	0	6	2	5	1	3	9	8
10	4	7	0	6	2	5	1	3	9	8

S

0	1	2	3	4	5	6	7	8	9
5	6	35	44	49	57	61	68	91	99

CLO – 6
Question No. 6

SO – A
(10) points

Write the pseudo-code / algorithm to print the given string character by character but in reverse.

Input: a “string” from user

e.g. “ALGORITHM”

Process: separate the characters of the “string”

Output: print character by character in reverse

“MHTIROGLA”

Solution:

Extra Sheet for rough work